

EE4437 HW2

Q1)

\* Use Infinite QW approx.

GaAs

 $\text{Al}_{0.2}\text{Ga}_{0.8}\text{As}$ 

$$\Rightarrow \Delta E_c = 0.77 \times 0.2 = 0.154 \text{ eV}$$

$$\Delta E_v = 0.48 \times 0.2 = 0.096 \text{ eV}$$

$$E_g = 1.424 + 1.27(0) = 1.424 \text{ eV}$$

particle masses:

$$m_e = (0.067 + 0.083x)m_0$$

since Infinite QW;  $x=0 \Rightarrow$  GaAs well

$$m_{hh} = (0.50 + 0.29x)m_0$$

$$m_e = 0.067 m_0$$

$$m_{lh} = (0.087 + 0.063x)m_0$$

$$\Rightarrow m_{hh} = 0.50 m_0$$

$$m_{lh} = 0.087 m_0$$

$$m_0 = 9.11 \times 10^{-31} \text{ kg}, \hbar = 1.055 \times 10^{-34} \text{ J}\cdot\text{s}, L_z = 10 \text{ nm}$$

$$E_{1e} = \frac{\hbar^2}{2m_e} \left(\frac{\pi}{L_z}\right)^2$$

$$= 0.056172 \text{ eV (5sf)}$$

$$\text{no } E_{2e} \text{ as } 2^2 \times 0.056172 \text{ eV} > 0.154 \text{ eV } (\Delta E_c)$$

$$E_{1hh} = \left(\frac{0.50}{0.067}\right) \frac{\hbar^2}{2m_e} \left(\frac{\pi}{L_z}\right)^2$$

$$= 0.0075270 \text{ eV (5sf)}$$

$$E_{2hh} = 2^2 \times E_{1hh} = 0.030108 \text{ (5sf)}$$

$$E_{3hh} = 3^2 \times E_{1hh} = 0.067743 \text{ eV (5sf)}$$

$$\text{no } E_{4hh} \text{ as } 4^2 \times E_{1hh} > \Delta E_v$$

$$E_{1lh} = \left(\frac{0.087}{0.067}\right) \frac{\hbar^2}{2m_e} \left(\frac{\pi}{L_z}\right)^2$$

$$= 0.043259 \text{ eV (5sf)}$$

$$\text{no } E_{2lh} \text{ as } 2^2 E_{1lh} > \Delta E_v$$

since only 1 electron energy level, lowest energy values determined by hole energy levels

$$E_{ph1} = E_{1e} + E_g + E_{1hh} = 1.4877 \text{ eV (5sf)}$$

$$E_{ph2} = E_{1e} + E_g + E_{2hh} = 1.51028 \text{ eV}$$

$$E_{ph3} = E_{1e} + E_g + E_{1lh} = 1.5234 \text{ eV (5sf)}$$

Q2)

$$a) E_F = \frac{\hbar^2}{2m^*} (3\pi^2 N)^{2/3}$$

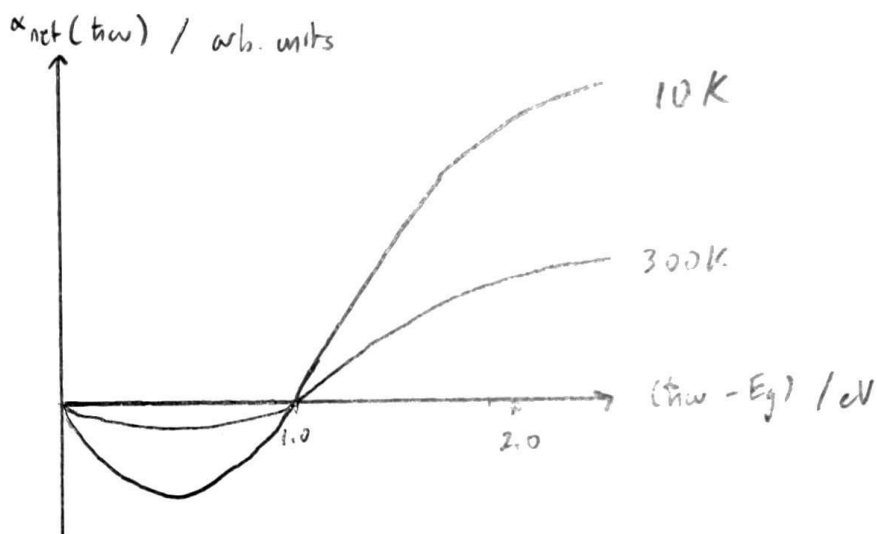
$N \text{ m}^{-3}$

$$E_{Fe} - E_c = \frac{\hbar^2}{2 \times 0.055 m_0} (3\pi^2 (5 \times 10^{25}))^{2/3} = 0.90056 \text{ eV (5sf)}$$

b)

$$E_v - E_{Fhh} = \frac{\hbar^2}{2 \times 0.5 m_0} (3\pi^2 (5 \times 10^{25}))^{2/3} = 0.099064 \text{ eV (5sf)}$$

c)



Between 10 K and 300 K,  $E_{Fe}$ ,  $E_{Fhh}$  remain constant

$$\alpha_{net}(h\nu) \propto (1 - (f_e + f_h)) ; f_e, f_h \propto T \Rightarrow \alpha_{net}(h\nu) \downarrow \text{ as } T \uparrow$$

d)

$$\text{Photon energy for which no net absorption} = (E_{Fe} - E_c) + E_g + (E_v - E_{Fhh})$$

$$\text{Width of gap region} = (E_{Fe} - E_c) + E_g + (E_v - E_{Fhh}) - E_g$$

$$= 0.99962 \text{ eV (5sf)}$$

$$\approx 1.00 \text{ eV}$$

Q2c)

$$E_V - E'_{Fhh} = \frac{\hbar^2}{2 \times 0.5 m_0} (3\pi^2 N')^{2/3}$$

$$= 0.15726 \text{ eV (SIF)}$$

⇒ width of gap region will increase

since width of gap region =  $(E_{Fe} - E_c) + (E_V - E_{Fhh})$ , since the second term ↑ and first term no change, sum ↑

⇒  $E_{Fhh} \uparrow$ , which means that  $\alpha_{net}(\hbar\omega) \uparrow$  since  $\alpha_{net}(\hbar\omega) \propto [1 - (f_e + f_h)]$

∴ net absorption spectrum will show larger gap region width, with larger values in y-axis.

Q3)

$$E_g = 1.519 \text{ eV}, m_e = 0.0665 m_0, m_{hh} = 0.45 m_0, m_0 = 9.11 \times 10^{-31} \text{ ✓}$$

$$E_{Fe} - E_c = \frac{\hbar^2}{2 \times m_e} (3\pi^2 (2 \times 10^{24}))^{2/3} = 0.087118 \text{ eV (SIF)}$$

$$E_V - E_{Fh} = \frac{\hbar^2}{2 \times m_{hh}} (3\pi^2 (2 \times 10^{24}))^{2/3} = 0.012874 \text{ eV (SIF)}$$

$$r_{\text{spont}}(\hbar\omega) = \frac{8\pi n^2}{h^3 c^2} [\hbar\omega]^2 f_e f_h \alpha_0(\hbar\omega)$$

$r_{\text{spont}}(\hbar\omega) / \text{arb. unit.}$

